

Signals

Signal.c

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    printf("\nSIGINT received!\n");
    printf("Signal number = %d\n", sig);
}

int main()
{
    signal(SIGINT, handler);

    while (1)
    {
        printf("Program is running...\n");
        sleep(2);
    }

    return 0;
}|
```

Output

```
ketana_sindhu@KetaZone:~$ nano signals.c
ketana_sindhu@KetaZone:~$ gcc signals.c -o s
ketana_sindhu@KetaZone:~$ ./s
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
^C
SIGINT received!
Signal number = 2
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
^Z
[1]+  Stopped                  ./s
ketana_sindhu@KetaZone:~$ |
```

Sigterm

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if (sig == SIGTERM)
        printf("\nSIGTERM received!\n");
}

int main()
{
    signal(SIGTERM, handler);

    printf("Process ID: %d\n", getpid());
    printf("Waiting for SIGTERM...\n");

    while (1)
    {
        sleep(2);
    }

    return 0;
}|
```

Output

```
ketana_sindhu@KetaZone:~$ nano SigTerm.c
ketana_sindhu@KetaZone:~$ gcc SigTerm.c -o ST
ketana_sindhu@KetaZone:~$ ./ST
Process ID: 935
Waiting for SIGTERM...
|
```

```
ketana_sindhu@KetaZone:~$ kill -SIGTERM 935
ketana_sindhu@KetaZone:~$ |
```

SigChild

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>

void child_handler(int sig)
{
    printf("SIGCHLD received: Child has terminated\n");
    wait(NULL);
}

int main()
{
    signal(SIGCHLD, child_handler);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child is running...\n");
        sleep(2);
        printf("Child is exiting...\n");
    }
    else
    {
        printf("Parent is waiting...\n");
        sleep(5);
        printf("Parent finished.\n");
    }

    return 0;
}
```

Output

```
ketana_sindhu@KetaZone:~$ nano SigChild.c
ketana_sindhu@KetaZone:~$ gcc SigChild.c -o SC
ketana_sindhu@KetaZone:~$ ./SC
Parent is waiting...
Child is running...
Child is exiting...
SIGCHLD received: Child has terminated
Parent finished.
ketana_sindhu@KetaZone:~$ |
```

SigPipe

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <stdlib.h>

void handle_sigpipe(int sig)
{
    printf("\nSIGPIPE received!\n");
    printf("The reading end of the pipe is closed.\n");
    exit(1);
}

int main()
{
    int fd[2];

    signal(SIGPIPE, handle_sigpipe);

    pipe(fd);

    printf("Pipe created.\n");

    // Close the reading end
    close(fd[0]);

    printf("Reading end closed.\n");

    printf("Trying to write to the pipe...\n");

    write(fd[1], "Hello", 5);

    printf("Write completed.\n");

    return 0;
}
```

Output

```
ketana_sindhu@KetaZone:~$ nano SigPipe.c
ketana_sindhu@KetaZone:~$ gcc SigPipe.c -o SP
ketana_sindhu@KetaZone:~$ ./SP
Pipe created.
Reading end closed.
Trying to write to the pipe...

SIGPIPE received!
The reading end of the pipe is closed.
ketana_sindhu@KetaZone:~$ |
```

Sigsegv

Code

```
GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <stdlib.h>

void handle_sigsegv(int sig)
{
    printf("\nSIGSEGV received!\n");
    printf("Invalid memory access detected.\n");
    exit(1);
}

int main()
{
    signal(SIGSEGV, handle_sigsegv);

    printf("Program started.\n");

    int *p = NULL;

    printf("Trying to access invalid memory...\n");

    *p = 10;

    printf("Program completed.\n");

    return 0;
}
```

Output

```
ketana_sindhu@KetaZone:~$ nano Sigsegv.c
ketana_sindhu@KetaZone:~$ gcc Sigsegv.c -o SS
ketana_sindhu@KetaZone:~$ ./SS
Program started.
Trying to access invalid memory...

SIGSEGV received!
Invalid memory access detected.
ketana_sindhu@KetaZone:~$ |
```